

# How to convert Gazebo SDF worlds to Webots?

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2 We are in the process of moving our warehouse simulation from Gazebo to Webots. We would like to set up a "realistic" warehouse environment in Webots. The factory sample already provides a lot of props for this, but we have not yet found warehouse shelf models for Webots.

One option would be to convert the AWS RoboMaker worlds from SDF to Webots:  
<https://github.com/aws-robotics/aws-robomaker-small-warehouse-world>

Is there a tool to make this conversion, or do we have to do it manually?

If no automated solution is available, what is the best approach to import the models manually?

webots

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asked Mar 29, 2023 at 16:29

 **Gergely Kis**  
23   2

## 1 Answer

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3 Although not automatic, it should be pretty straightforward to re-use the shelf object from this AWS RoboMaker world by simply re-using its meshes to create a `Shelf.proto` file in your local `protos` folder:

 

```
#VRML_SIM R2023a utf8
# A shelf.

PROTO Shelf [
  field SFVec3f    translation 0 0 0
  field SFRotation rotation  0 0 1 0
  field SFString   name       "shelf"
  field SFBool     locked     FALSE           # Is `Solid.locked`.
]
{
  Solid {
    translation IS translation
    rotation IS rotation
    children [
      CadShape {
        url "https://raw.githubusercontent.com/aws-robotics/aws-robomaker-
```

```

small-warehouse-
world/ros1/models/aws_robomaker_warehouse_ShelfD_01/meshes/aws_robomaker_warehouse
    }
    ]
    name IS name
    model "shelf"
    boundingObject Mesh {
        url "https://raw.githubusercontent.com/aws-robotics/aws-robomaker-small-
warehouse-
world/ros1/models/aws_robomaker_warehouse_ShelfD_01/meshes/aws_robomaker_warehouse
    }
    locked IS locked
    }
}

```

Here is what it looks like in Webots:



Here is the sample.wbt world file showing it in Webots:

```

#VRML_SIM R2023a utf8

EXTERNPROTO
"webots://projects/objects/backgrounds/protos/TexturedBackground.proto"
EXTERNPROTO
"webots://projects/objects/backgrounds/protos/TexturedBackgroundLight.proto"
EXTERNPROTO "../protos/Shelf.proto"

WorldInfo {
}
Viewpoint {
    orientation -0.11384659601330234 -0.3187749829101217 0.9409683644240309
5.557715208500585
    position -5.776529177279546 4.724710358949437 2.980453565305262
}
TexturedBackground {
}
TexturedBackgroundLight {
}
Shelf {
    rotation 0 0 1 1.5708003061004252
}

```

You can also view it in 3D online on [webots.cloud](https://webots.cloud).

**Note:** Using meshes for the bounding object is not optimal, especially in this case where the bounding object could be replaced with a [compound](#) object made up of simple boxes, e.g., using a `Group` node that contains a series of `Transform` nodes containing a `Box` node in their `Shape` node.

*Disclaimer: I am a [Webots](#) developer working at [Cyberbotics](#).*

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edited Mar 30, 2023 at 8:29

answered Mar 30, 2023 at 7:44

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Olivier Michel

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